MDPI

Article

WRF Cold-Pool Structure and Evolution during a Uinta Basin Winter Ozone Episode

John R. Lawson [1,2,*], Michael J. Davies [1], Loknath Dhar [1,3] and Seth N. Lyman [1,3]

[1] Bingham Research Center, Utah State University, Vernal, UT 84078, USA;
[2] Department of Mathematics and Statistics, Utah State University, Logan, UT 84322, USA;
[3] Department of Chemistry and Biochemistry, Utah State University, Logan, UT 84322, USA
* Correspondence: john.lawson@usu.edu

Abstract

Cold-pool structure controls the mixing volume and transport available to winter-ozone chemistry, but surface skill alone does not establish that structure. We evaluated Weather Research and Forecasting simulations of the 2 February 2013 Uinta Basin episode, anchored by four runs that crossed GFS and NAM driving products with one-way and two-way nest feedback. Across five post-spin-up hours, the GFS members were warmer at the surface, about 3.6 K less stable from 50 to 500 m above ground, and 0.8–1.0 m s^{-1} faster in fixed-layer transport wind than the matched NAM members. They also had less footprint-integrated heat deficit, but every simulation gained heat deficit during the window; the product separation was a state offset, not faster GFS cold-pool loss. Nest-feedback effects were much smaller, and the boundary-layer/surface-layer treatment produced a smaller, metric-dependent response. The design does not isolate initial from lateral-boundary forcing, validate inversion depth or event-scale persistence, or predict ozone.

Keywords: winter ozone; WRF; cold-air pool; Uinta Basin; driving data; nest feedback; heat deficit; model evaluation

1. Introduction

[Diagnostic — not gate-passed: Working-draft mode. This revision is deliberately broad but selective: it carries every archived controlled comparison and each candidate result thread with a scoped evidential role, while leaving redundant or unreadable figure galleries in the reproducible harvest. Three status classes appear throughout. *Promoted* figures and *confirmed* claims have passed the repository's provenance and admission gates and carry ledger identifiers (C-numbers). *Registered candidates* are tracked with full provenance but are not admitted. *Exploratory renders* are batch engine figures catalogued in `figures-draft/HARVEST.md`; they illustrate visual leads only, and no quantitative claim in this draft rests on them. The analysis gates named in this draft (ANA-17, ANA-18, ANA-19, ANA-20) have all been executed with criteria fixed in committed scripts before any output was read; each section reports its gate's pass or fail outcome, and the complete decision record is `rewrite/HYBRID-GATE-REVIEW.md`. A submission-ready revision would remove the remaining diagnostic scaffolding and admit only figures that pass the normal promotion path.]

Winter ozone in the Uinta Basin develops when photochemistry proceeds within snow-covered, stably stratified air that restricts vertical dilution and horizontal exchange [1–3]. Snow increases actinic flux and modifies deposition, while the stable, weakly ventilated

Received:
Revised:
Accepted:
Published:

Copyright: © 2026 by the authors. Submitted to *Air* for possible open access publication under the terms and conditions of the Creative Commons Attribution (CC BY) license.

air confines ozone and its precursors near the basin floor. For a coupled model, the meteorological question is therefore not only whether 2 m temperature has a small mean error. It is whether the atmosphere delivered to the chemistry grid contains a credible shallow mixing environment and weak transport during the accumulation period.

Published concentration agreement does not provide a clean ranking of the relevant uncertainties. Ahmadov et al. obtained a much stronger ozone simulation after applying an aircraft-derived, Horsepool-centered emissions scaling while holding the meteorological simulation fixed, and they conditioned the resulting inventory interpretation on adequate meteorology [1]. Matichuk et al. evaluated a cumulative ladder that changed volatile-organic-compound emissions, leaf area, and deposition together, while improved WRF temperature profiles did not consistently improve ozone profiles [4]. These outcomes are consistent with compensating error among confinement, emissions, photolysis, deposition, and chemical mechanism. They make meteorological evaluation necessary for chemical attribution without making meteorology the uniquely dominant error source.

The meteorological literature also warns against reducing an experiment to a configuration label. Neemann et al. showed strong case-specific responses from bundled snow, albedo, and cloud-microphysics choices, but their ice-retention modification was a bounding treatment and the 1.33 km grid evaluated most closely was finer than the 4 km meteorology supplied to the chemistry model [2]. Tran et al. showed that the meteorological response to four-dimensional data assimilation depended on the nudged variables, input data, and diagnosed boundary-layer depth; their disclosed artificial formaldehyde limits quantitative chemical inference [5]. Thus, labels such as "NAM driven," "nudged," or "high resolution" do not by themselves identify a controlled cause, and aggregate scores can obscure a failure to retain the layer of interest.

The archive is not a chronological experiment sequence. It is a graph of simulations connected by controlled contrasts, and we organize it by the lever that each contrast can identify. The first question is whether the short-window lower-layer state responds reproducibly to the GFS-versus-NAM driving-data product under both nest-feedback modes. The second is how the two process treatments represented in the fine-terrain branch—slope-aware radiation and a coupled boundary-layer/surface-layer scheme change—alter that state. The third is whether the within-product feedback response remains small at both default and fine terrain. The fourth is whether changing only the static-terrain source produces a coherent basin-floor or terrain-organized response. For the driving-product contrasts we additionally ask whether the response appears in a footprint-integrated heat-deficit measure and whether the products differ in its within-window change.

Contrasts that share the frozen station and common-height pipeline are displayed together before any individual result is developed. Terrain-source contrasts instead require area-weighted geometry diagnostics, because a station-only summary can miss a rim- or slope-confined response. The stable X0–X10 and CMP01–CMP12 codes are retained for provenance, but run numbers do not set manuscript order or evidential weight. The heat-deficit comparison carries two criteria that were fixed before any output was inspected, and all executed gates retain their pass or fail outcome. The design evaluates the meteorological input state that a later chemistry calculation would receive; it does not contain a chemistry calculation.

2. Materials and Methods

2.1. Case, Observations, and Evaluation Domains

The target is the snow-covered Uinta Basin cold-air pool on 2 February 2013 within the episode examined by prior meteorological and chemical studies [1,2]. All common-sample contrasts use 14:00–18:00 UTC, corresponding to 07:00–11:00 mountain standard

time. The first two forecast hours were excluded as spin-up, leaving five hourly samples; the sensitivity of the results to that two-hour choice has not been evaluated. Surface observations were extracted from the Synoptic archive for eleven stations reporting 2 m temperature, ten reporting 10 m wind speed, and nine reporting surface pressure in every retained hour. Subhourly reports were averaged within each UTC clock hour. The stations span basin-floor, sidewall, and bench locations but are neither spatially uniform nor area weighted. Every controlled simulation discussed below archived both the 1 km d02 and the 333 m d03 during this window, so output availability did not determine the evaluation domain. The 1 km d02 was selected for the common observation-linked estimand because it contains all eleven stations, whereas seven stations clamp to the boundary of the roughly 50 by 48 km d03 and only four provide usable interior columns there (ledger row C03). Figure 1 temporarily shows the domain geometry and reporting stations. This coverage decision does not establish that d02 resolves every process response adequately or that a response measured on d02 has the same magnitude or ordering on d03. The present results therefore describe the full-network d02 sample; a d03 analysis would require a separately declared interior footprint or the same four-station subset on both domains and is not a resolution evaluation. Four elevation-spanning stations provide surface context from 31 January 00:00 UTC through 4 February 00:00 UTC. Observed and modeled surface potential temperature for a two-site structural index was calculated as $\theta = T(1000/p)^{0.2854}$, with T in kelvin and p in hPa, and the index was defined as Mountain Home on the northwest bench minus Pariette Draw on the basin floor. Because the two sites are horizontally separated and locally exposed, the index is a spatial surface contrast rather than a colocated free-air inversion strength or depth. No sounding, profiler, or remote-sensing profile is used to validate the modeled 50–500 m layer.

[Rewrite gate: Replace Figure 1 with a two-panel orientation figure before coauthor circulation. Panel (a) should locate the observation sites against named towns, major roads, rivers, the basin floor, and surrounding mountain ranges; panel (b) should show the common d01–d02–d03 nesting. Remove all current and proposed transects because no retained result uses or explains them, and suppress latitude/longitude axes in favor of a north arrow and scale bar.]

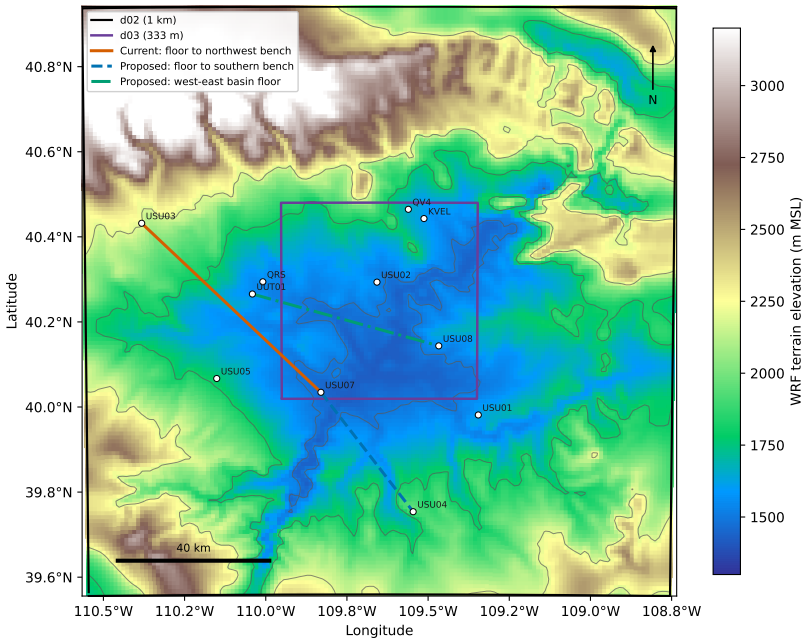


Figure 1. Temporary locator showing the reporting stations on d02 and the d03 interior domain. The overlaid transects are not used in this manuscript and will be removed when the two-panel geographic and domain locator is regenerated. Seven of the eleven reporting stations fall outside the usable d03 interior, which is why d02 supplies the full-network common sample; the choice does not validate d02 against d03 (C03).

[Rewrite gate: Synoptic citation wording, the current API defaults, access conditions, and the frozen-cache hash are documented. Before circulation, archive or regenerate the historical request with explicit quality-control parameters, station and network ownership, sensor metadata, completeness screening, uncertainty treatment, request URL, and provider-specific reuse permission.]

2.2. WRF Configuration and Archived Simulation Family

The controlled six-hour simulations used WRF version 4.8.0 from 12:00 to 18:00 UTC on 2 February 2013. They used three nested domains with horizontal grid spacing of 3, 1, and 0.333 km, 75 eta levels, and a 50 hPa model top. The reference physics suite used Thompson microphysics, RRTMG longwave and shortwave radiation, the revised MM5 surface layer, the Noah land-surface model, and the YSU planetary-boundary-layer scheme. Grid and observation four-dimensional data assimilation were disabled. All deliberate departures from that reference configuration are listed by simulation in Table 1. The GFS one-way member reused the six verified metgrid files from the GFS two-way member before rerunning `real.exe` and `wrf.exe`.

[Rewrite gate: The WRF Version 4 technical reference and v4.8.0 release, GFS grid 004, and NAM grid 218 identities are verified. Insert only the keys generated by John's small curated bibliography export, and recover the exact input filenames and hashes, map projection, domain dimensions and centers, time step, boundary interval, static-geography versions, and snow-field initialization from the archived as-run records.]

The archive contains eleven stable simulation identifiers, X0–X10 (Table 1). An X code identifies a simulation configuration; it is not itself an experiment or a claim. Run identity, exact archive paths, namelist fingerprints, and feedback flags are recorded in the run inventory, and directory or figure labels are never used as identity. As-run differencing connects the controlled family into four scientific branches: driving-data product, boundary-layer and radiation treatments, nest feedback, and static-terrain source. The X7 and X8

configurations differ from the controlled family in several dimensions at once and enter only as design context (Section 2.4).

Table 1. The archived simulation family. Every run uses WRF 4.8.0, three nests at 3/1/0.333 km with 75 levels, Thompson microphysics, RRTMG radiation, and Noah land surface unless stated. "Fine terrain" is a 3-arc-second static source, "default" a 30-arc-second-class source, and "coarse" a 5-arc-minute source; source resolution is not model-grid spacing. X7 and X8 are context only: they differ from the controls in several dimensions at once.

Code	Product, feedback	Deliberate configuration role	Status
X0	NAM, one-way	fine-terrain control, YSU/MM5	control for X1, X3, X4, X9
X1	NAM, one-way	X0 plus slope radiation and shading	controlled pair (CMP07)
X2	NAM, one-way	X1 with MYJ/Eta instead of YSU/MM5	controlled pair (CMP08)
X3	NAM, one-way	default terrain; reference configuration	pairs CMP02, 03, 09, 10
X4	NAM, one-way	deliberately coarse 5-arc-minute terrain	lineage closed; pairs CMP10, 11
X5	NAM, two-way	default terrain	pairs CMP01, 03
X6	GFS, two-way	default terrain	pairs CMP01, 04
X7	NAM, two-way	1 km inner grid, 24 h integration	context only (CMP12)
X8	NAM, one-way	four domains to 111 m, 50 levels	context only (CMP12)
X9	NAM, two-way	X0 with two-way feedback	controlled pair (CMP06)
X10	GFS, one-way	default terrain	pairs CMP02, 04

2.3. Controlled Contrasts and Estimands

A CMP code identifies a contrast between simulations and therefore the lever that the analysis can estimate (Table 2). The table is ordered by scientific role rather than identifier so that archive chronology does not determine manuscript prominence. The four default-terrain simulations X3, X5, X6, and X10 form a complete two-by-two driving-product/feedback matrix, hereafter the default-terrain matrix. The products supplied zero-hour analyses at 12:00 and 18:00 UTC and provided 27 and 40 metgrid levels for GFS and NAM, respectively. The GFS-versus-NAM treatment is therefore a product-level bundle of native grids, analysis systems, vertical input, assimilated observations, the 12:00 UTC initial state, and subsequent lateral-boundary evolution. The design cannot assign a product response to one of those components. Within each product family, the normalized as-run namelists differ only in the feedback flag. The two driving-product contrasts are evaluated separately so that their response must recur under both feedback settings, while the feedback contrasts and paired interaction serve as robustness checks rather than a coequal thesis. The remaining controlled contrasts enter on their own design and evidence: the X2-minus-X1 scheme treatment, X1-minus-X0 slope-radiation treatment, X9-minus-X0 fine-terrain feedback treatment, and the pairwise X3/X0/X4 terrain-source brackets.

Table 2. Controlled comparisons referenced in this draft, their authoritative differencing orientation, and current status. "Confirmed" means a ledgered claim supports scoped prose now; "analyzed" means the matched recomputation is complete and committed; "diagnostic" means figures exist but no gate-passed quantitative claim; "context" means the comparison is not controlled.

ID	Operation	Isolated lever	Status and gate
CMP01	X6 − X5	driving product (two-way)	confirmed (C04, C17, C25); spatial gate executed and failed (C30)
CMP02	X10 − X3	driving product (one-way)	confirmed (C04, C17, C25); spatial gate executed and failed (C30)
CMP08	X2 − X1	PBL and surface-layer scheme	confirmed station contrast (C32, refining C19); sole C6 candidate screen executed and failed on a measurement-limited criterion (C31)
CMP07	X1 − X0	slope radiation and shading	confirmed near-null (C18, C25)
CMP03	X3 − X5	feedback, NAM, default terrain	confirmed small (C18, C25)
CMP04	X10 − X6	feedback, GFS, default terrain	gate passed; confirmed bound (C28; Section 3.7)
CMP06	X9 − X0	feedback, NAM, fine terrain	confirmed small (C18, C25)
CMP05	(X10−X6)−(X3−X5)	driving-product-by-feedback interaction	gate executed and failed (ANA-17; gate review)
CMP09	X3 − X0	terrain source, default vs fine	controlled after lineage closure (C29); diagnostics committed, no gate declared
CMP10	X3 − X4	terrain source, default vs coarse	controlled after lineage closure (C29); diagnostics committed, no gate declared
CMP11	X0 − X4	terrain source, fine vs coarse	confirmed negative design finding (C16)
CMP12	X7 / X0 / X8	none (multiply confounded)	

2.4. Design Exclusions: Resolution and Legacy Context

The archive contains runs whose innermost grids span nominal 1 km, 333 m, and 111 m spacing (X7, X0, and X8), and a visually tempting "resolution ladder" could be drawn from them. That ladder would be false, and this is a confirmed design finding (C16): the three configurations differ simultaneously in initialization and forecast age, domain count and placement, feedback mode, vertical levels, terrain treatment, radiation options, and output cadence (Table 3). No grid-spacing response can be attributed within this set, and nominal fine spacing is not evidence of a better cold-pool simulation. A controlled resolution experiment would require a new matched family; X7 and X8 therefore do not enter the Results comparison hierarchy. Legacy nudged and SIP-era simulations

are likewise heterogeneous in episode, model version, and assimilation state and remain context only.

Table 3. Why X7/X0/X8 is not a resolution experiment: dimensions on which the members of the apparent ladder differ simultaneously (C16). Any one row would already confound a grid-spacing attribution.

Dimension	Differences across X7 / X0 / X8
Innermost nominal spacing	1 km / 333 m / 111 m
Domain count and placement	different nest counts, extents, and centers
Integration length and age	24 h integration for X7 against 6 h for X0 and X8
Feedback mode	two-way (X7) against one-way (X0, X8)
Vertical levels	75 (X7, X0) against 50 (X8)
Terrain treatment	default-class against 3-arc-second static sources and placements
Radiation options	slope-aware options differ across the set
Output cadence	hourly output windows differ with integration design

2.5. Surface and Common-AGL Diagnostics

Modeled 2 m temperature, earth-relative 10 m wind, and surface pressure were sampled at the nearest d02 cell to each station. Across the eleven fixed columns, model terrain minus station elevation had a median of -2.0 m and ranged from -32.4 to +14.3 m. That agreement limits one obvious mismatch but does not remove exposure, slope, land-cover, or network-coverage error. Mass-level height above ground was calculated from perturbation and base-state geopotential and the local WRF terrain height. Grid-relative winds were destaggered and rotated to earth-relative components. Potential temperature and both wind components were interpolated linearly at 25 m increments from 50 to 500 m above ground level (AGL) at each station column. Fixed-layer stability was defined as $S_{50:500} = \theta(500\,\mathrm{m}) - \theta(50\,\mathrm{m})$. The transport-wind proxy was $U_{50:500} = \{\bar{u}^2 + \bar{v}^2\}^{1/2}$, where the bars denote height means from 50 to 500 m AGL. This wind metric is not flow through the basin boundary, pollutant residence time, cold-pool volume, or a chemistry-model ventilation coefficient. Native WRF planetary-boundary-layer (PBL) height was retained only as a scheme-dependent companion diagnostic.

2.6. Footprint-Integrated Heat Deficit

The original basin-volume heat deficit of Whiteman et al. represents the sensible heat required to remove a basin inversion, and Adler et al. provide the closest published column implementation to the present diagnostic. **[Rewrite gate: Insert the Paperpile-generated keys for Whiteman et al. (1999) and Adler et al. (2023) after John's small curated export; do not invent or hand-edit keys.]** The present area-normalized atmospheric heat-deficit proxy was defined as $H = (c_p/g) \int \max\{\theta_{\mathrm{crest}} - \theta, 0\}\, dp$, integrated from the surface to a fixed crest reference height, where θ_{crest} is the potential temperature interpolated to that height in each modeled column. H is reported as a cell-area-weighted mean over an evaluation footprint in MJ m^{-2}. The footprint is the convex hull of the eleven evaluation stations and eleven registered basin waypoints, intersected with model terrain below the crest reference. The primary definition uses a 2200 m crest and the combined hull, covering 4965 d02 cells and 5277 km^2. Six alternate definitions vary the crest to 2100 and 2300 m, dilate or erode the boundary by one cell, or restrict the hull to the stations or waypoints alone. Because two-way feedback modifies the d02 terrain field by up to 23.1 m relative to the static one-way field, differences were formed within feedback mode, X6 minus X5 and X10 minus X3, on a common mask fixed from the one-way NAM terrain; the members of each pair share identical grids. Hourly values at 13:00 UTC provide trajectory context only, and all decisions use 14:00–18:00 UTC. Two criteria were fixed before inspection:

a weaker-state criterion requiring both GFS-minus-NAM pairs to be negative at every retained hour under all seven definitions, and a faster-loss criterion requiring both GFS 14:00-to-18:00 UTC changes to be negative and more negative than the matched NAM changes under all seven definitions. The fixed 2200 m crest, nonnegative clipping, convex-hull footprint, and modeled profiles depart from the published definitions. The footprint is an evaluation region rather than a hydrologic basin boundary, and the proxy is neither a direct reproduction of the published diagnostics nor an observed cold-pool measurement.

3. Results

3.1. Observed Surface Context

[Rewrite gate: Add an early two-panel episode-context figure before the surface-temperature series. Panel (a) should show a verified, minimally cloudy MODIS true-color or snow product from 1–3 February 2013 over the Uinta Basin; panel (b) should show quality-controlled observed ozone for January–February 2013 with 2 February marked by a red line and annotation. State the satellite product, overpass time, processing level, ozone sites, averaging rule, units, missingness, and data access provenance; use the figure as chemical and snow context, not as validation of modeled vertical structure.]

The basin-floor station at Pariette Draw repeatedly cooled below the higher sites from 31 January through 3 February, while the magnitude and ordering of the station temperatures changed through the diurnal cycle (Figure 2). The recurring floor-to-bench separation is consistent with cold-pool conditions around the modeled period. It does not determine the cold-pool top, depth, volume, or vertical temperature structure.

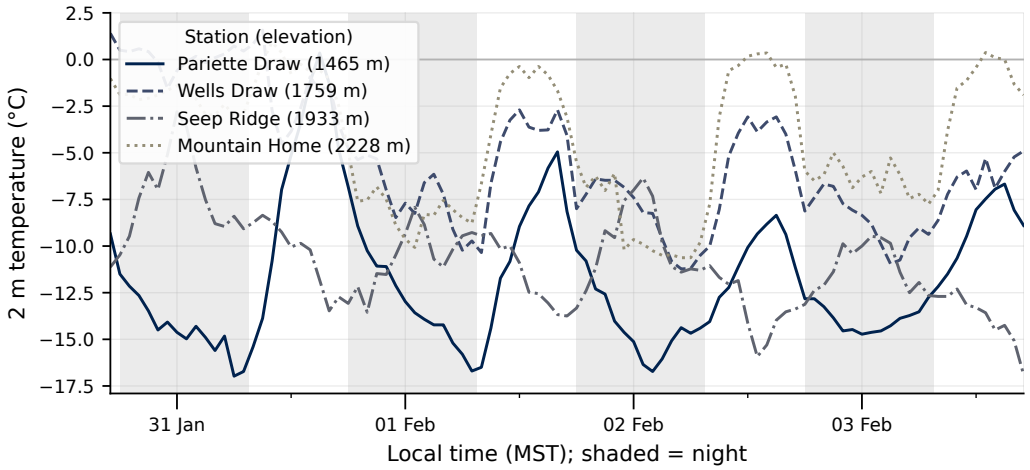


Figure 2. Observed 2 m-temperature evolution at four elevation-spanning Uinta Basin stations from 31 January through 3 February 2013. Shading indicates approximate local night. The sites establish a recurring spatial surface-temperature pattern but do not constitute a colocated vertical profile.

3.2. Common-Pipeline Contrasts at a Glance

Figure 3 answers one question: which controlled changes mattered most when every comparison used the same station and common-AGL pipeline? The driving-product contrasts dominate the temperature, fixed-layer stability, and transport-wind panels. The PBL/surface-layer treatment is smaller but detectable, whereas the slope-radiation, nest-feedback, and interaction effects cluster near zero on the same scales. Pressure and wind error do not reproduce that ordering, so the figure is a response hierarchy rather than a ranking of the best configuration.

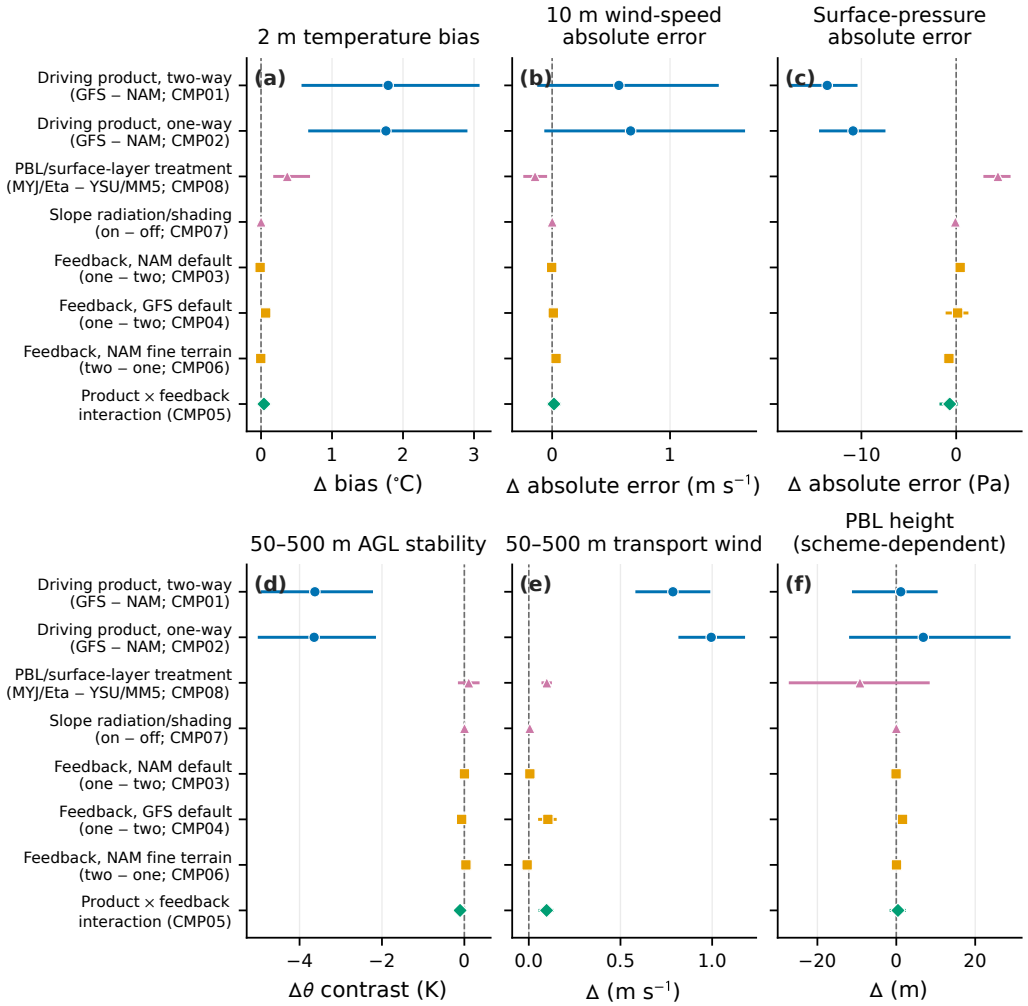


Figure 3. Mean five-hour effects for the controlled comparisons that share the station/common-AGL pipeline. Colors and marker shapes identify driving-product, process-treatment, feedback, and interaction contrasts; each row states its differencing direction. Whiskers are descriptive block-sensitivity intervals for five dependent hourly medians, not confidence intervals. Working figure drawn from committed summaries; not yet registered or promoted.

3.3. Driving-Product Response in the Common Lower Layer

The common-AGL diagnostics separated the driving products under both feedback settings (Figure 4). GFS minus NAM reduced 50–500 m potential-temperature stability by 3.63 K under two-way feedback and 3.65 K under one-way feedback (Table A1). The five hourly spatial-median differences ranged from -5.58 to -1.13 K and from -5.36 to -0.90 K, respectively. All ten product-by-hour stability contrasts therefore retained the same sign. GFS minus NAM increased fixed-layer transport wind by 0.79 m s^{-1} under two-way feedback and 0.99 m s^{-1} under one-way feedback. The hourly differences ranged from +0.37 to +1.18 m s^{-1} and from +0.61 to +1.50 m s^{-1}, with the same sign in every hour. Native PBL-height differences changed sign within both feedback modes and did not reproduce the stability and wind separation.

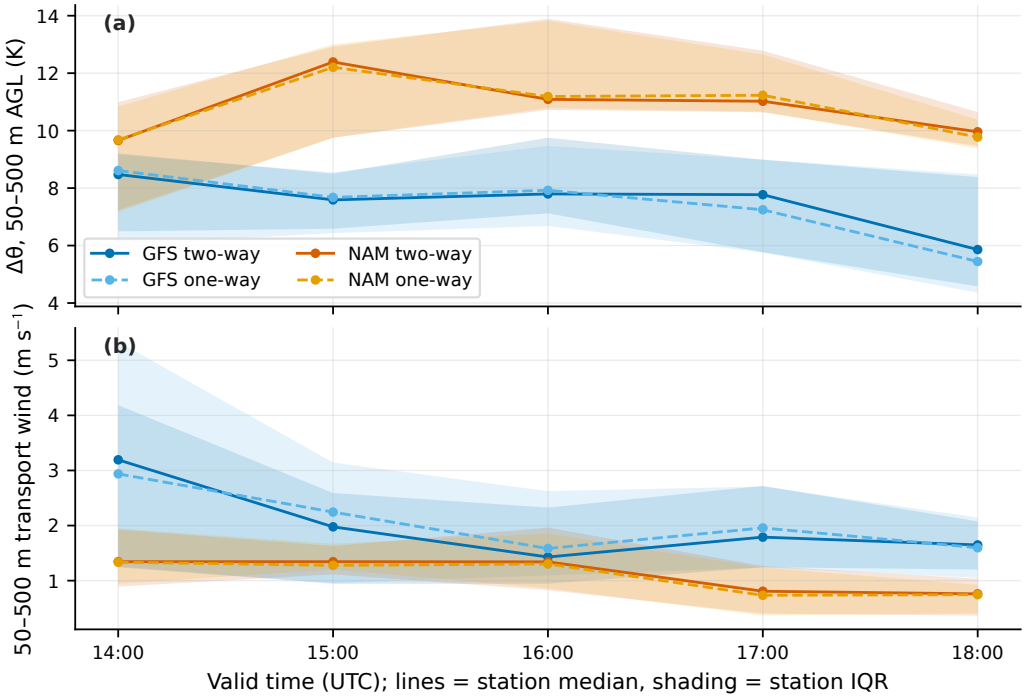


Figure 4. Station-column medians and interquartile ranges for 50–500 m AGL potential-temperature stability and fixed-layer transport wind over the matched window. Solid lines denote two-way feedback and dashed lines denote one-way feedback. The lower-layer GFS–NAM separation recurs under both feedback modes.

3.4. Surface and Spatial Corroboration

The paired surface-temperature result had the same product ordering as the layer diagnostics.

The paired GFS-minus-NAM temperature-bias contrasts were +1.79 and +1.76 °C, but surface-pressure absolute error changed in the opposite direction and hourly wind-error differences changed sign (Table A4). No single surface score therefore ranks the products. The observed Mountain Home-minus-Pariette Draw potential-temperature contrast supplies useful surface context, but the two stations are horizontally separated and do not measure free-air inversion depth.

Across the registered footprint, the mean GFS-minus-NAM near-surface response was positive and the vertical response remained concentrated below 500 m AGL under both feedback modes (Table A5). The predeclared spatial gate nevertheless failed because only 0.64–0.67 of the footprint shared the median sign at 14:00–15:00 UTC, just below the two-thirds criterion. The result is therefore shallow and mean-coherent, but not spatially uniform throughout the retained window. Figures 5 and 6 are restored in this review draft because they show the spatial patchiness hidden by the footprint mean and the repeated vertical structure under both feedback settings. They do not reverse the failed early-hour coherence gate or promote the unexplained elevated response into a new claim.

[Rewrite gate: Figures 5 and 6 are registered diagnostics restored for editorial review, not promoted manuscript figures. Before coauthor circulation, decide whether each belongs in the main paper, supplement, or reviewer reserve and regenerate any retained version under the current figure-production contract.]

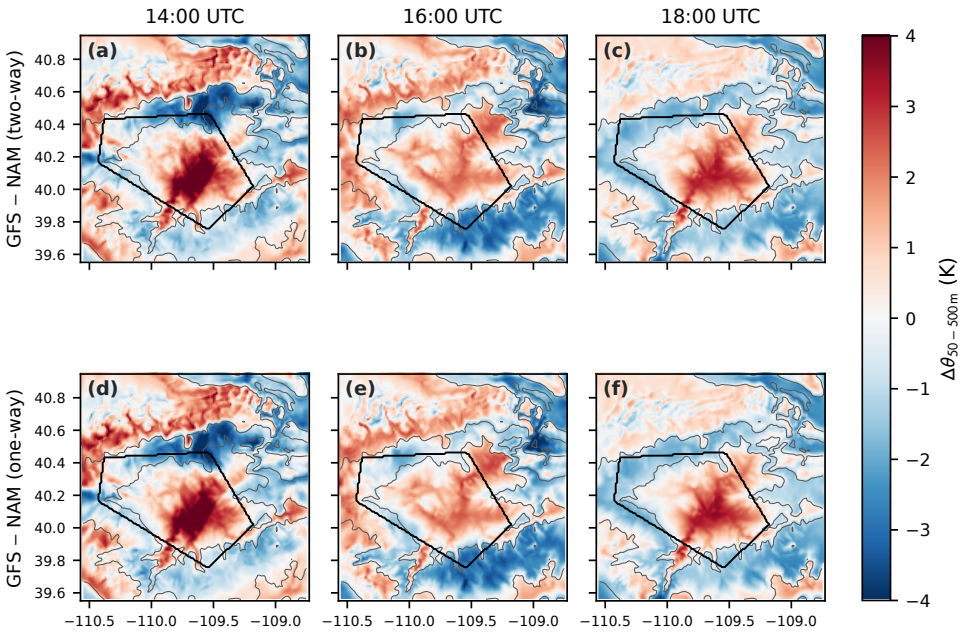


Figure 5. GFS-minus-NAM 50–500 m AGL potential-temperature response on d02 at 14:00, 16:00, and 18:00 UTC under two-way feedback (top) and one-way feedback (bottom). The black outline is the registered evaluation footprint. The response becomes more spatially coherent after the two early hours that failed the predeclared sign-share criterion, but the later pattern cannot rescue that failed gate.

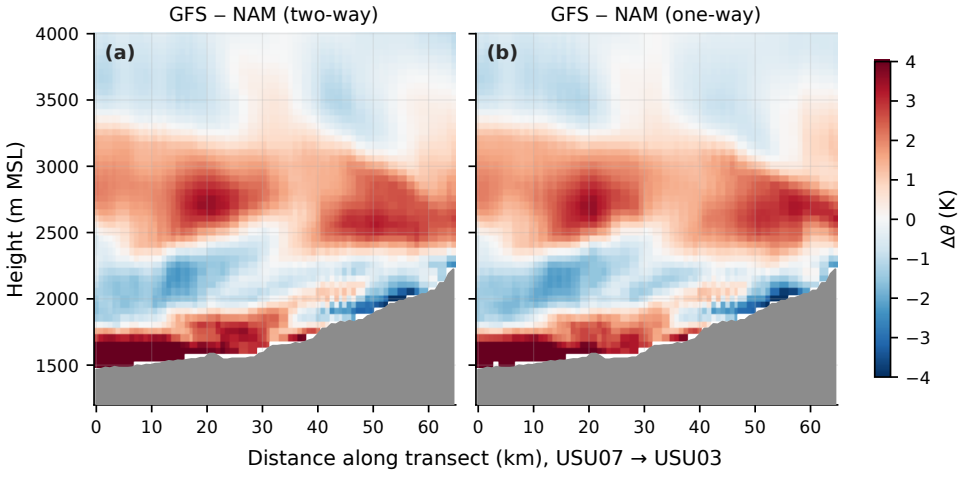


Figure 6. GFS-minus-NAM potential-temperature difference at 16:00 UTC along the Pariette Draw-to-Mountain Home transect under two-way feedback (left) and one-way feedback (right). Both panels show a warmer GFS lower layer beneath a cooler layer and a separate elevated positive response. The elevated feature is retained as diagnostic structure and is not assigned a mechanism in this manuscript.

3.5. Footprint Heat Deficit: A Weaker State without Faster Loss

The GFS members carried less footprint-integrated heat deficit at every retained hour under every tested footprint definition (Table A2). The mean GFS-minus-NAM differences were -1.23 and -1.25 MJ m^{-2} under two-way and one-way feedback. Heat deficit increased in all four simulations, however, so the predeclared faster-loss criterion failed under every definition. This is evidence of a state offset already present at 14:00 UTC, not of different cold-pool evolution during the five-hour window. The restored trajectory in Figure 7 makes

that distinction visible, while Figure 8 shows that the 16:00 UTC offset is spatially structured rather than a uniform basin-wide decrement.

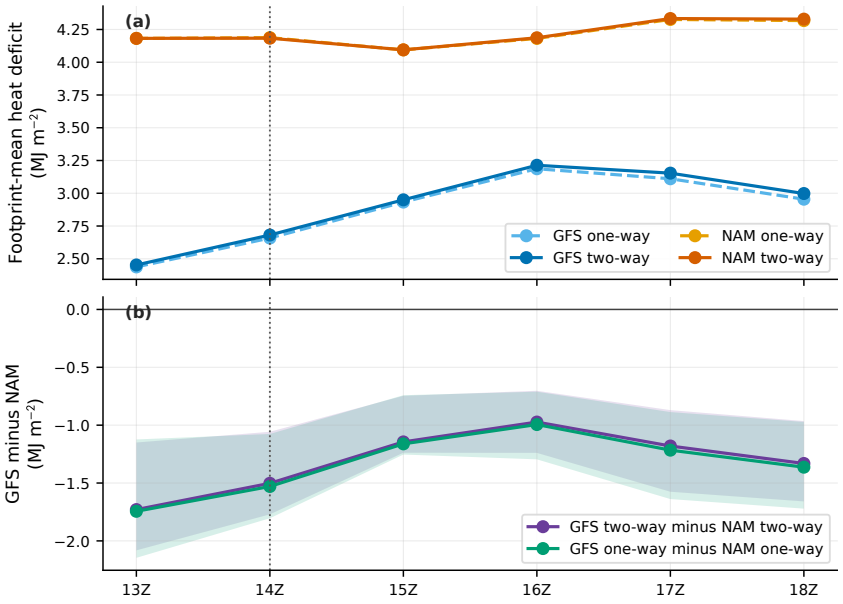


Figure 7. Footprint-mean heat-deficit trajectories for the four default-terrain matrix members and seven-definition sensitivity envelopes for the two GFS-minus-NAM pairs. The 13:00 UTC values are context only; all criteria use 14:00–18:00 UTC. GFS retains a smaller deficit throughout, but the deficit does not decline faster in either GFS member.

[Rewrite gate: Figure 8 is an exploratory external-engine render restored for editorial review. Its source family and sign are recorded in `figures-draft/HARVEST.md`, but it lacks a per-image generator sidecar and must be registered, restyled, and admitted before circulation.]

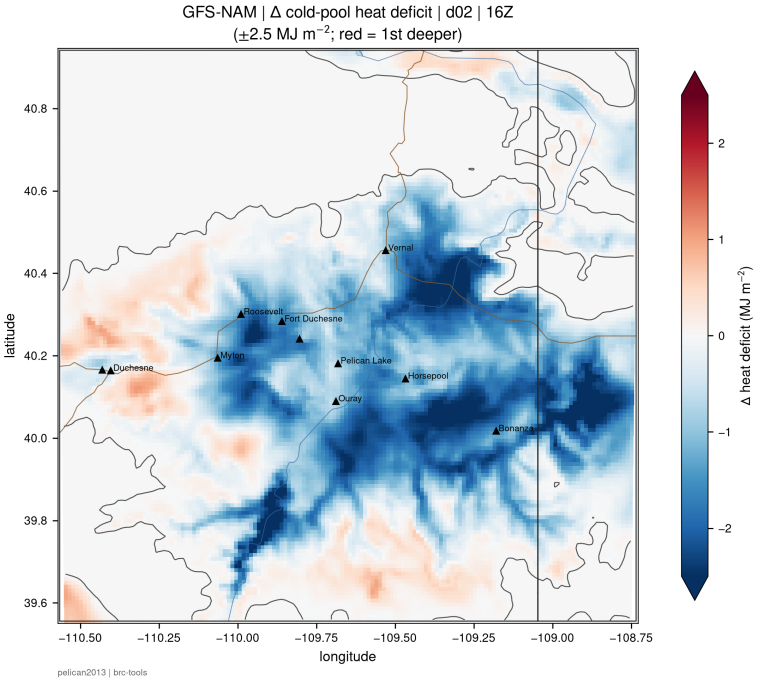


Figure 8. Exploratory plan-view GFS-minus-NAM cold-pool heat-deficit difference on d02 at 16:00 UTC for the two-way pair. Negative values across much of the basin floor correspond to the smaller GFS heat deficit summarized in Figure 7. This single snapshot illustrates spatial structure; it cannot establish erosion, persistence, or basin export.

3.6. Process Treatments on the Fine-Terrain Branch

The scheme pair (CMP08, X2−X1) changes only the boundary-layer and surface-layer schemes, MYJ/Eta against YSU/MM5, while holding the fine terrain and slope-aware radiation configuration fixed. MYJ/Eta increased 2 m-temperature bias by 0.37 °C, reduced 10 m-wind absolute error by 0.145 m s^{-1}, and increased fixed-layer transport wind by 0.098 m s^{-1}, with the same sign in all five hours (Table A3; C32). These changes are smaller than the driving-product response on the shared lower-layer metrics, and the opposing temperature- and wind-error changes do not support a scheme ranking. PBL height is scheme defined and is not compared as a common physical layer.

The slope-radiation treatment produced essentially no change at the basin-floor station columns (C31). An aspect-organized skin-temperature response appeared on the sidewalls, but the archived flat-plane shortwave field and missing surface heat fluxes could not test the mechanism directly. The predeclared screen therefore failed for a measurement-limited reason. The diagnostic is retained in Appendix A (Figure A1); it does not support a main-text claim that slope radiation improved or degraded the cold pool.

3.7. Nest-Feedback Robustness across Two Terrain Regimes

Within-product feedback effects were small beside the repeated GFS-minus-NAM separation on the default-terrain matrix (Table A1). The fine-terrain pair reproduced that bound: stability changed by +0.04 K and transport wind by -0.01 m s^{-1}, while their sensitivity intervals spanned zero (Table A6; C28). Small, consistently signed wind-error and pressure changes do not support a configuration recommendation. Nest feedback therefore serves as a replicated robustness check for this case and window, not as a principal result or a claim of general equivalence.

3.8. Terrain-Source Brackets (CMP09–CMP11)

The archive brackets the static terrain source on an otherwise identical NAM one-way configuration: default minus fine (CMP09, X3−X0), default minus coarse (CMP10, X3−X4), and fine minus coarse (CMP11, X0−X4). The default-minus-fine terrain change was small on d02, and its 0.07–0.27 K response failed the predeclared terrain-correlation criterion early in the window (Table A7; C29). The coarse-terrain brackets produced larger responses, but no criterion was declared for them before inspection. They remain diagnostic rather than evidence for a terrain ranking. The four-panel terrain figure is omitted because its near-null panels consume more space than the failed-gate result warrants; the registered render remains archived.

4. Discussion

4.1. The Main Result Is a Difference in the Supplied Lower-Layer State

The clearest result is not that one simulation wins a surface-score contest. It is that the GFS-driven members supplied a warmer, less stable, and faster-moving 50–500 m layer than the matched NAM members under both feedback settings. The footprint heat-deficit calculation supports the same ordering beyond the fixed station columns. The mapped and section diagnostics in Figures 5 and 6 show both the repeated lower-layer structure and the spatial qualifications hidden by an aggregate value. Replication across feedback makes the product separation robust to that configuration choice within this matrix, while the opposite pressure-error ordering prevents a universal product ranking.

This distinction matters to an air-quality model because meteorology sets the layer in which ozone and its precursors mix and move. The GFS state is more consistent with dilution than the matched NAM state, but the transport-wind proxy is not basin export and the heat-deficit footprint is not an observed mixing volume. No chemical species were

simulated, so the result identifies a meteorological precondition for attribution rather than an ozone response. That boundary is necessary because emissions, snow and cloud treatment, deposition, chemical mechanism, and meteorological confinement can compensate for one another in coupled simulations [1,2,4].

The driving-product treatment also bundles the 12:00 UTC initial state with later boundary updates. The present design cannot assign the lower-layer response to one of those components. "Driving-product response" is therefore warranted; "initial-condition effect" is not.

4.2. The Difference Is a State Offset, Not Demonstrated Cold-Pool Loss

The heat-deficit result narrows the interpretation. GFS had less deficit than NAM under every feedback and footprint definition, but deficit increased in every simulation during the retained window (Figure 7). The predeclared faster-loss test therefore failed, and the product separation must be described as a state offset already present by 14:00 UTC. A longer integration through at least one complete diurnal cycle is required to test whether the products differ in cold-pool erosion or persistence.

The spatial analysis gives a similarly bounded result. The product response was strongest below 500 m AGL and had the expected footprint-mean sign at every hour, but the two-thirds sign-coherence criterion failed at 14:00–15:00 UTC. The response is shallow and mean-coherent on d02, not spatially uniform throughout the window. Preserving these failed criteria is more informative than rescuing them with a new threshold after inspection.

4.3. Secondary Experiments Narrow Rather than Expand the Claim

The process and geometry comparisons are useful chiefly because they delimit the main result. The common-pipeline overview in Figure 3 makes that response hierarchy explicit on the shared metrics. The boundary-layer/surface-layer treatment produced a coherent but smaller response, with temperature error worsening while wind error improved. It is a process-treatment contrast, not evidence that one named scheme is generally superior. Nest-feedback effects remained small at both default and fine terrain, so feedback protects the product result as a robustness check without becoming a second thesis.

The slope-radiation and fine-terrain screens did not earn main-text figures. Slope radiation was nearly invisible at the basin-floor stations, while its sidewall skin-temperature signal could not be tested against a slope-corrected radiative or surface-flux diagnostic. The fine-terrain response was small and failed its predeclared correlation criterion; the larger coarse-terrain response had no predeclared gate. These outcomes motivate better measurements and future experiment design, but they do not support radiation or terrain rankings. The multiply confounded X7/X0/X8 set is excluded for the same reason: nominal grid spacing cannot be isolated from its other configuration changes.

4.4. Observation and Design Limits Set the Next Experiment

The common-AGL result leads because it applies one geometric layer, station sample, time window, and aggregation to all four anchor simulations. Native PBL height cannot replace it because PBL height is scheme dependent and its product contrasts changed sign. The observed floor-to-bench temperature contrast is valuable surface corroboration, but horizontally separated stations do not validate inversion depth, layer stability, or modeled heat deficit. Observed ozone and satellite snow cover will strengthen the episode context, not supply that missing vertical validation or establish a meteorology–ozone causal response.

The d02 choice is likewise a sampling decision rather than a resolution result. It retains all eleven stations, whereas d03 retains only four usable interior columns, but it

may smooth slope, tributary, and terrain-edge responses. A predeclared transfer test must compare d02 and d03 over the same geographic support and the same four-station sample before the present response hierarchy is assumed to hold at 333 m.

The five-hour window, fixed two-hour spin-up exclusion, dependent hourly summaries, and incomplete observation provenance limit every conclusion. The next discriminating experiment should combine a longer matched product/feedback matrix with an independent in-basin vertical reference and a design that separates initial from lateral-boundary forcing. Only then should selected meteorological states be crossed with emissions and chemical mechanisms to test an ozone response. All failed gates in the present archive remain failed until new evidence is generated under a newly declared criterion.

5. Conclusions

The largest repeated response in the controlled archive is the GFS-versus-NAM driving-product contrast on the common d02 evaluation sample. Relative to NAM, GFS is warmer at the evaluated surface sites, about 3.6 K less stable from 50 to 500 m AGL, 0.8–1.0 m s^{-1} faster in fixed-layer transport wind, and lower in footprint-integrated heat deficit under both feedback modes. The heat-deficit separation is a state offset, not demonstrated faster loss, and the surface metrics do not support a universal product ranking. Nest-feedback effects are much smaller, the PBL/surface-layer response is smaller and metric dependent, and the slope-radiation and fine-terrain screens do not pass their declared gates. The result establishes a five-hour difference in the meteorological state supplied to downstream chemistry; it does not identify the responsible product component, validate inversion depth, establish event-scale persistence, calculate basin export, or predict ozone. The next experiment requires a longer matched matrix, independent vertical observations, a common-support d02–d03 transfer test, and a design that separates initial from lateral-boundary forcing.

Appendix A. Data Tables and Secondary Diagnostics

This appendix retains the numerical detail and failed-gate diagnostics needed to audit the concise Results narrative.

Table A1. Paired effects for the default-terrain driving-product/feedback matrix, averaged over the five hourly spatial medians. Positive temperature values indicate a warmer model-minus-observation bias, positive stability values indicate a more stable fixed layer, and positive transport-wind values indicate a larger layer-mean wind. The effects are deterministic descriptive contrasts rather than inferential estimates.

Effect	ΔT2 bias (°C)	$\Delta S_{50:500}$ (K)	$\Delta U_{50:500}$ (m s^{-1})
GFS − NAM, two-way	+1.79	-3.63	+0.79
GFS − NAM, one-way	+1.76	-3.65	+0.99
GFS one-way − two-way	+0.07	-0.07	+0.10
NAM one-way − two-way	-0.01	+0.00	+0.01
Feedback interaction	+0.04	-0.10	+0.10

Table A2. Footprint-integrated heat-deficit contrasts over the 14:00–18:00 UTC decision window. The primary definition uses the combined station-and-waypoint hull with a 2200 m crest (4965 cells, 5277 km^2); envelopes span all seven footprint definitions. The values are deterministic descriptive contrasts in MJ m^{-2}.

GFS − NAM quantity	Two-way pair	One-way pair
Mean difference (primary)	-1.23	-1.25
Hourly range (primary)	-1.50 to -0.97	-1.53 to -0.99
Hourly envelope (seven definitions)	-1.77 to -0.70	-1.80 to -0.71
GFS change, 14:00–18:00 UTC	+0.32	+0.30
NAM change, 14:00–18:00 UTC	+0.15	+0.13
Difference in change (primary)	+0.17	+0.17
Weaker-state criterion passed	7/7	7/7
Faster-loss criterion passed	0/7	0/7

Table A3. Executed process-pair station contrasts on the frozen pipeline (14:00–18:00 UTC; job 13948378). Values are means of five hourly spatial medians; sign counts are positive/negative hours with exact zeros omitted. PBL height is scheme defined across CMP08 and is retained only with that caveat.

Metric	CMP07 (X1 − X0)		CMP08 (X2 − X1)	
	Mean	Signs	Mean	Signs
T2 bias (°C)	+0.001	3/1	+0.37	5/0
10 m wind absolute error (m s^{-1})	+0.001	2/2	-0.145	0/5
Surface-pressure bias (Pa)	-0.12	0/4	+6.3	5/0
Stability $S_{50:500}$ (K)	+0.004	3/1	+0.11	3/2
Transport wind $U_{50:500}$ (m s^{-1})	+0.006	4/0	+0.098	5/0
PBL height (m)	+0.01	3/1	-9.2	2/3

Table A4. Surface corroboration for the driving-product contrasts. Every value uses the matched station and hour samples described in Section 3.4.

Evidence	Result	Interpretation limit
2 m temperature	GFS-minus-NAM bias contrast +1.79 °C two-way and +1.76 °C one-way; absolute-error differences +1.41 and +1.50 °C	Same ordering as common-AGL stability
10 m wind	Bias contrast +0.91 and +0.96 m s^{-1}; hourly absolute-error differences change sign	No stable wind-skill ordering
Surface pressure	Absolute-error contrast -13.6 Pa two-way and -10.9 Pa one-way; negative in every hour	Opposite ordering from temperature, with small absolute differences
Two-site surface index	Observed 12.63–20.21 K; NAM 16.44–18.07 K; GFS 7.35–11.74 K under two-way feedback	Horizontally separated surface contrast, not inversion strength or depth

Table A5. Spatial and vertical checks for the two GFS-minus-NAM driving-product contrasts over the registered d02 footprint (ANA-19; C30).

Quantity	Committed range	Criterion and outcome
Mean 2 m potential-temperature difference	+0.89 to +1.16 K at 14:00–15:00 UTC; +2.00 to +2.74 K at 16:00–18:00 UTC	Positive at all ten pair-hours
Footprint sign share	0.64–0.67 at 14:00–15:00 UTC; 0.84–0.93 later	Two-thirds criterion failed at the first two hours
Low-to-aloft absolute-response ratio	1.45–3.12	Greater than one at all ten pair-hours
50–500 m stability difference	-4.33 to -3.04 K	Negative at all ten pair-hours
Gate conjunction	Two of three component criteria passed throughout	Failed because early floor sign coherence missed its threshold

Table A6. Fine-terrain feedback contrast CMP06 (X9 minus X0, two-way minus one-way). Values are means of five hourly spatial medians from the committed recomputation (job 13948333); sensitivity intervals are descriptive, not confidence intervals.

Metric	Mean	Hourly range	Signs (+/−)	Sens. interval
Stability $S_{50:500}$ (K)	+0.04	-0.09 to +0.14	4/1	-0.03 to +0.10
Transport wind $U_{50:500}$ (m s^{-1})	-0.01	-0.07 to +0.03	3/2	-0.03 to +0.01
PBL height (m)	+0.05	-0.67 to +0.78	3/2	-0.38 to +0.47
T2 bias (°C)	-0.004	-0.09 to +0.04	3/2	-0.05 to +0.03
T2 absolute error (°C)	-0.006	-0.06 to +0.05	2/3	-0.04 to +0.03
10 m wind bias (m s^{-1})	+0.02	-0.01 to +0.06	4/1	+0.01 to +0.04
10 m wind absolute error (m s^{-1})	+0.03	+0.001 to +0.06	5/0	+0.01 to +0.06
Surface-pressure bias (Pa)	-0.77	-1.23 to -0.35	0/5	-1.06 to -0.50
Surface-pressure absolute error (Pa)	-0.77	-1.23 to -0.42	0/5	-1.01 to -0.59

Table A7. Terrain-source diagnostics retained after the failed fine-terrain gate and the closed X4 lineage check (ANA-17; C29).

Contrast	Terrain difference	2 m response	Status
CMP09, default minus fine	7.9 m RMS; -55 to +60 m range	Elevation-band means 0.07–0.27 K; terrain correlation below 0.3 at 14:00–16:00 UTC	Predeclared conjunction failed
CMP10/CMP11, coarse brackets	91.9/93.1 m RMS; extrema to -462/+487 m	Slope-band means 1.2–1.5 K; terrain correlation 0.50–0.67	Descriptive only; no predeclared gate
X4 lineage	As-run namelist matches the controlled family; terrain is materially smoother	Configuration caveat closed	Supports controlled-bracket status, not a terrain ranking

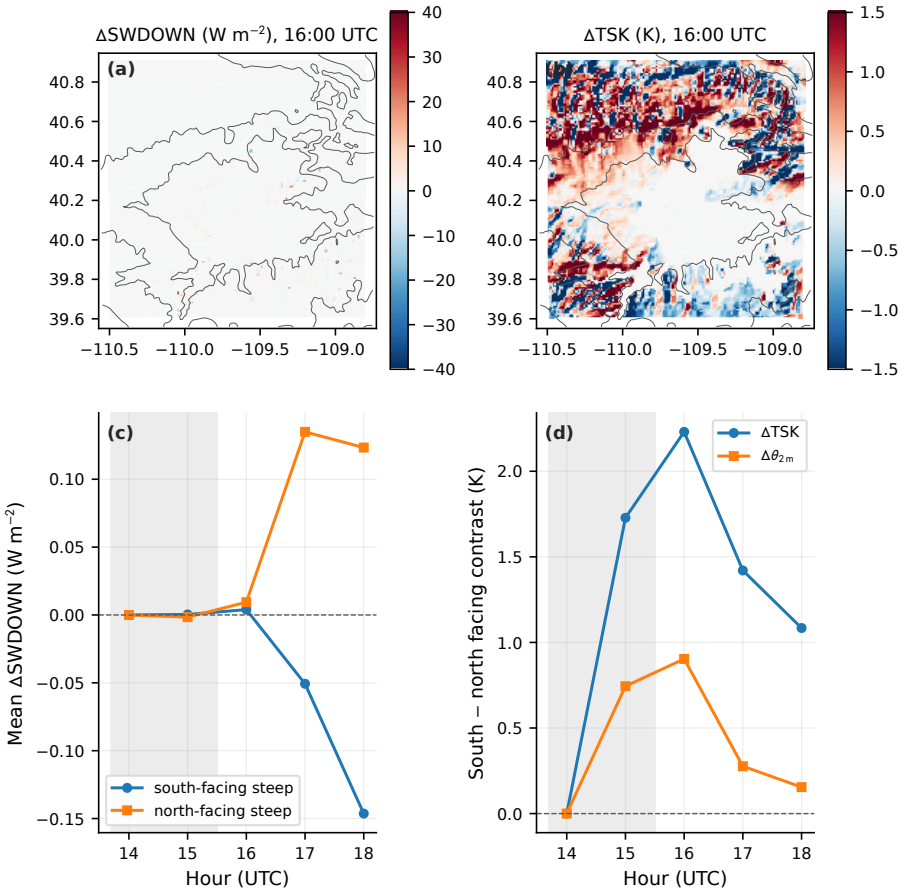


Figure A1. Diagnostic for the slope-radiation screen (ANA-18; C31). The archived flat-plane shortwave field did not express the treatment, while skin temperature showed an aspect-organized response on steep terrain. The predeclared screen failed for this measurement-limited reason, and the figure supports no configuration ranking.

Author Contributions: [Rewrite gate: Assign and approve CRediT roles for all authors.]

Funding: [Rewrite gate: Insert verified funders and grant numbers.]

Institutional Review Board Statement: Not applicable.

Informed Consent Statement: Not applicable.

Data Availability Statement: The analysis scripts, derived hourly and paired summaries, heat-deficit tables, run and comparison inventories, provenance records, and promoted figures used for this draft are retained in the private working repository. The exploratory renders reproduced in this working revision are catalogued with source paths and status in `figures-draft/HARVEST.md`. **[Rewrite gate: Add a public repository or DOI, licenses and access conditions for raw WRF and Synoptic data, and the exact release commit before submission.]**

Acknowledgments: [Rewrite gate: Identify computational facilities, data providers, and contributors who do not meet the authorship criteria.]

Conflicts of Interest: [Rewrite gate: All authors must confirm the conflicts-of-interest statement.]

Use of Artificial Intelligence: Generative artificial intelligence tools, including Anthropic Claude and OpenAI Codex, were used during manuscript preparation to assist with literature synthesis, provenance and analysis-code review, and drafting and editing. The authors reviewed and edited the resulting material and take responsibility for the content.

References

1. Ahmadov, R.; McKeen, S.; Trainer, M.; Banta, R.; Brewer, A.; Brown, S.; Edwards, P.M.; De Gouw, J.A.; Frost, G.J.; Gilman, J.; et al. Understanding high wintertime ozone pollution events in an oil-and natural gas-producing region of the western US. *Atmos. Chem. Phys.* **2015**, *15*, 411–429.
2. Neemann, E.M.; Crosman, E.T.; Horel, J.D.; Avey, L. Simulations of a cold-air pool associated with elevated wintertime ozone in the Uintah Basin, Utah. *Atmos. Chem. Phys.* **2015**, *15*, 135–151. https://doi.org/10.5194/acp-15-135-2015.
3. Davies, M.J.; Lawson, J.R.; O'Neil, T.; Lyman, S.N.; Zager, K.; Coxson, T.D. Uinta Basin snow shadow: Impact of snow-depth variation on winter ozone formation. *Air (Basel)* **2025**, *3*, 22. https://doi.org/10.3390/air3030022.
4. Matichuk, R.; Tonnesen, G.; Luecken, D.; Gilliam, R.; Napelenok, S.L.; Baker, K.R.; Schwede, D.; Murphy, B.; Helmig, D.; Lyman, S.N.; et al. Evaluation of the Community Multiscale Air Quality Model for Simulating Winter Ozone Formation in the Uinta Basin. *J. Geophys. Res. D: Atmos.* **2017**, *122*, 13545–13572. https://doi.org/10.1002/2017JD027057.
5. Tran, T.; Tran, H.; Mansfield, M.; Lyman, S.; Crosman, E. Four dimensional data assimilation (FDDA) impacts on WRF performance in simulating inversion layer structure and distributions of CMAQ-simulated winter ozone concentrations in Uintah Basin. *Atmos. Environ. (1994)* **2018**, *177*, 75–92. https://doi.org/10.1016/j.atmosenv.2018.01.012.

Disclaimer/Publisher's Note: The statements, opinions and data contained in all publications are solely those of the individual author(s) and contributor(s) and not of MDPI and/or the editor(s). MDPI and/or the editor(s) disclaim responsibility for any injury to people or property resulting from any ideas, methods, instructions or products referred to in the content.